

RCDEPS Drive Systems — CDR Addendum v1.0

Detailed pod architecture, EM design, vectoring, controls, EMC, thermal, and test plans

1. Scope & Objectives

This addendum freezes the propulsion-plant details for a purpose-built RCDEPS ship: pod architecture, ring integration, vectoring, power electronics, EM geometry, shielding, thermal rejection, control loops, fault management, and verification.

2. RCDEPS Pod Architecture (Block Diagram)

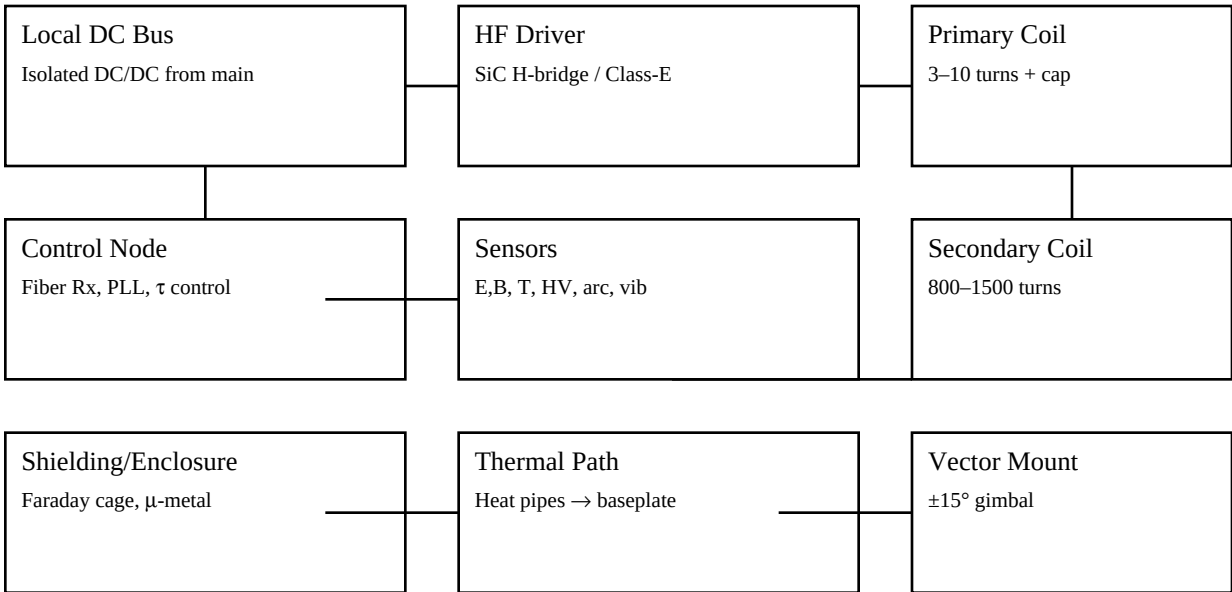


Figure 1. Single-Pod functional blocks and interfaces

3. Electromagnetic Geometry & Materials

Coils: Secondary $\varnothing$ 100–150 mm, $L=250\text{--}400$ mm, 800–1500 turns (AWG 28–34). Primary 3–10 turns litz with HV tank cap tuned to F_{res} ; toroidal/spherical topload $r_{edge} \geq 5$ mm to suppress field emission.

Dielectrics/supports: PTFE, PEEK, alumina standoffs; vented; no trapped volumes. Clearances ≥ 10 mm/kV along vacuum-facing paths.

Grounds: Pod Faraday cage to base; ship star ground; differential IO. Field shaping with guard rings and smooth fairings.

4. HF Driver & Power Electronics

Topology: SiC H-bridge or Class-E/F at 10–500 kHz; soft-switching to reduce EMI. Isolated DC/DC (1–5 kW) per pod; fiber commands only.

Protection: Desat/OC trips, dv/dt clamps, snubbers, crowbar on DC input; analog first. Filters: CM chokes, RC/LC; twisted/coax harnesses.

5. Vectoring Mount & Structure

Gimbal $\pm 15^\circ$ dual-axis; ceramic/hybrid bearings; nonmagnetic alloys. Harmonic drives; absolute encoders; manual detent.

No HV slip-rings; drivers local. Thermal: heat pipes to baseplate; straps to radiator rails.

6. Controls: β – ϕ – τ Loops & Thrust Aggregation

Per pod: track amplitude β and phase ϕ ; τ provides recursive feedback. Ring: vector-sum control; vehicle: MPC 10–20 Hz; attitude 20–50 Hz.

Watchdogs assert SAFE: $\beta \rightarrow 0$, ϕ hold, τ reset. Health: E-field, temp, vibration, arc detect, leakage current $\rightarrow$ 2oo3 voter.

7. EMC/EMI Cleanliness

Faraday-caged pods; RF gaskets; μ -metal around magnetics. One-point ground; surge/MOVs. Radiated emissions tests in-vacuo.

8. Thermal Design & Heat Rejection

Per pod dissipation 0.5–2 kW; conduction to baseplate; louvered radiators. Limits: Cu<120°C, diel<150°C, semis<100°C case.

9. Health Monitoring & Safing

States: OFF $\rightarrow$ STANDBY $\rightarrow$ RAMP $\rightarrow$ TRACK $\rightarrow$ THRUST $\rightarrow$ HOLD $\rightarrow$ SAFE. Trips force SAFE ($\beta=0$, crowbar, gimbal detent).

10. Verification Plan

Bench resonance/Q, corona check, hipot; vacuum thrust with photon calibrator; phase/ τ sweeps; EMI scans; 100 h endurance; HIL vectoring tests; fault injections.

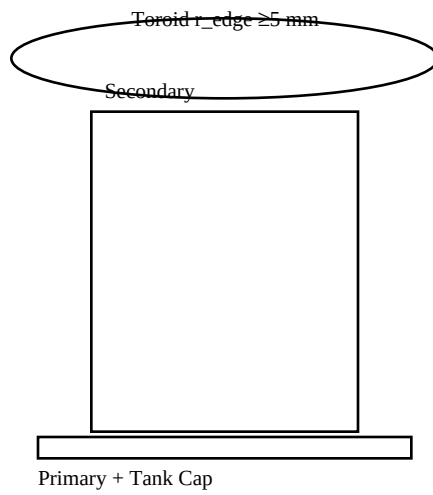
11. Targets & Scaling

Goal: $F \propto A \cdot V^2 \cdot \eta_{\text{coh}} \cdot f_{\text{res}}$; normalize to photon benchmark $F_{\text{ph}}=P/c$. Bands: 10–500 kHz RF with low-Hz envelope. 2 $\times$ HV clearance; 3 $\times$ thermal; 20 dB EMI margin.

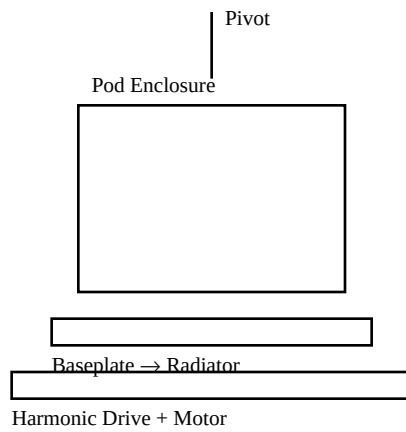
12. Pod BOM (abridged)

Subsystem	Key Parts
Coils/EM	Secondary coil form & wire, primary litz, HV tank cap, topload, PTFE/PEEK supports
Driver/Power	SiC devices, gate drivers, DC/DC isolator, snubbers, LC filters, crowbar, shunts
Control	Fiber transceiver, PLL, MCU/FPGA, analog watchdogs, sensors
Structure	Gimbal yoke, bearings, harmonic drive, encoders, baseplate
Thermal	Heat pipes, graphite pads, radiator brackets, NTCs/RTDs
Shielding	Faraday panels, RF gaskets, μ -metal, cable glands
Safety	Interlocks, E-stop, desat/OC detect, arc UV sensor, burst disks

Appendix A — Coil/Topload Cross-Section (Notional)



Appendix B — Vectoring Gimbal (Notional)



— End of CDR Addendum v1.0 —